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IEEE Recommended Practice for

Protection and Coordination of Industrial and Commercial Power Systems

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IEEE Recommended Practice for Protection and Coordination of Industrial and Commercial Power Systems

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Abstract: The principles of system protection and the proper selection, application, and coordination of components that may be required to protect industrial and commercial power systems against abnormalities that could reasonably be expected to occur in the course of system operation are presented in a in a simple, yet comprehensive, format. The principles presented apply to both new electrical system design and to the changing, upgrading, or expansion of an existing electrical distribution system.
Keywords: bus protection, cable protection, calibration, conductor protection, coordinating time intervals, current-limiting fuses, current transformers, fuse coordination, fuse selectivity, generator grounding, generator protection, high-voltage fuses, liquid preservation systems, low-voltage motor protection, medium-voltage motor protection, motor protection, overcurrent protection, potential transformers, power fuses, protective relays, relay application principles, relay operating principles, service protection, short-circuit protection, switchgear protection, system design, system protection, transformer protection, voltage transformers

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Introduction

(This introduction is not a part of IEEE Std 242-2001, IEEE Recommended Practice for Protection and Coordination of Industrial and Commercial Power Systems.)

IEEE Std 242-2001, the *IEEE Buff Book*™, has been extensively revised and updated since it was first published in 1975. The *IEEE Buff Book* deals with the proper selection, application, and coordination of the components that constitute system protection for industrial plants and commercial buildings. System protection and coordination serve to minimize damage to a system and its components in order to limit the extent and duration of any service interruption occurring on any portion of the system.

A valuable, comprehensive sourcebook for use at the system design stage as well as in modifying existing operations, the *IEEE Buff Book* is arranged in a convenient step-by-step format. It presents complete information on protection and coordination principles designed to protect industrial and commercial power systems against any abnormalities that could reasonably be expected to occur in the course of system operation.

Design features are provided for

— Quick isolation of the affected portion of the system while maintaining normal operation elsewhere
— Reduction of the short-circuit current to minimize damage to the system, its components, and the utilization equipment it supplies
— Provision of alternate circuits, automatic throwovers, and automatic reclosing devices

Participants

At the time this recomended practice was approved, the IEEE Working Group on Protection and Coordination of Industrial and Commercial Power Systems had the following members and contributors:

Carey Cook, *Chair*

Chapter 1: First principles—**Robert G. Hoerauf**, *Chair*; R. G. (Jerry) Irvine

Chapter 2: Short-circuit calculations—**Louie J. Powell**, *Chair*; David S. Baker, Chet Davis, John F. Witte

Chapter 3: Instrument transformers—**James D. Bailey** and **David S. Baker,** *Cochairs;* B. J. Behroon, Louie J. Powell, Marcelo Valdes

Chapter 4: Selection and application of protective relays—**Keith R. Cooper**, **N. T. (Terry) Stringer**, **John F. Witte**, *Cochairs*; Bruce G. Bailey, David S. Baker, James W. Brosnahan, Carey J. Cook, David D. Shipp, John F. Witte

Chapter 5: Low-voltage fuses—**Vincent J. Saporita**, *Chair*; Robert L. Smith, Alan F. Wilkinson

Chapter 6: High-voltage fuses—**John R. Cooper**, *Chair*; Carey J. Cook, Herb Pflanz, Kris Ranjan, John F. Witte

Chapter 7: Low-voltage circuit breakers—**George D. Gregory**, *Chair*; Bruce G. Bailey, John Chiloyan, Keith R. Cooper, R. G. (Jerry) Irvine, Steve Schaffer

Chapter 8: Ground-fault protection—**Shaun P. Slattery**, *Chair*; James P. Brosnahan, John Chiloyan, Edward Gaylon, Daniel. J. Love, Elliot Rappaport, Steven Schaffer, S. I. Venugopalan

Chapter 9: Conductor protection—**William Reardon**, *Chair*; John Chiloyan, George D. Gregory, Alan C. Pierce, Vincent J. Saporita

Chapter 10: Motor protection—**Daniel J. Love**, *Chair*; Al Hughes, Alan C. Pierce, Lorraine K. Padden, Joseph S. Dudor, R. G. (Jerry) Irvine

Chapter 11: Transformer protection—**Alan C. Pierce**, *Chair*; Carey J. Cook, John R. Cooper, Jerry Frank, Vincent J. Saporita, N. T. (Terry) Stringer, Ralph H. Young

Chapter 12: Generator protection—**Alan C. Pierce**, *Chairs*; Jay D. Fisher, Robert G. Hoerauf, Danial J. Love, Robert L. Simpson, Ralph H. Young

Chapter 13: Bus and switchgear protection—**Robert L. Smith, Jr**., *Chair*; Edward Gaylon, Steve Schaffer, John Steele

Chapter 14: Service supply line protection—**Lorraine K. Padden**, *Chair*; Consumer Interface Protection and Relaying Practices Subcommittees of the IEEE Power Systems Relaying Committee

Chapter 15: Overcurrent coordination—**N. T. (Terry) Stringer**, *Chair*; Bruce G. Bailey, Keith R. Cooper, Joseph S. Dudor, Douglas Durand, Tim Fink, Jay D. Fisher, George D. Gregory, William M. Hall, Steve Schaffer, John F. Witte, Ralph H. Young

Chapter 16: Maintenance, testing, and calibration—**R. G. (Jerry) Irvine**, *Chair*; Donald J. Akers, Jerry S. Baskin, Roderic L. Hageman, Ted Olsen, Gabe Paoletti, Elliot Rappaport, David, L. Swindler, Neil H. Woodley

The following members of the balloting group voted on this standard. Balloters may have voted for approval, disapproval, or abstention:

Bruce G. Bailey
James D. Bailey
David S. Baker
Ray M. Clark
Carey J. Cook
John R. Cooper
Chet Davis
George D. Gregory
Robert G. Hoerauf
R. G. (Jerry) Irvine
Daniel J. Love
Lorraine K. Padden
Alan C. Pierce
Louie J. Powell
William Reardon
Vincent J. Saporita
David D. Shipp
Shaun P. Slattery
Robert L. Smith, Jr.
N. T. (Terry) Stringer
John F. Witte
Ralph H. Young

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IEEE Recommended Practice for Protection and Coordination of Industrial and Commercial Power Systems

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First principles

1.1 Overview

1.1.1 Scope

IEEE Std 242-2001, commonly known as the *IEEE Buff Book*™ , is published by the Institute of Electrical and Electronics Engineers (IEEE) as a reference source to provide a better understanding of the purpose for and techniques involved in the protection and coordination of industrial and commercial power systems.

IEEE Std 242-2001 has been prepared on a voluntary basis by engineers and designers functioning as a working group within the IEEE, under the Industrial and Commercial Power Systems Department of the Industry Applications Society. This recommended practice is not intended as a replacement for the many excellent texts available in this field. IEEE Std 242-2001 complements the other standards in the *IEEE Color Book Series*™ , and it emphasizes up-to-date techniques in power system protection and coordination that are most applicable to industrial and commercial power systems. Coverage is limited to system protection and coordination as it pertains to system design treated in IEEE Std 141-1993[1] and IEEE Std 241-1990. No attempt is made to cover utility systems or residential systems, although much of the material presented is applicable to these systems.

This publication presents in a step-by-step, simplified, yet comprehensive, form the principles of system protection and the proper application and coordination of those components that may be required to protect industrial and commercial power systems against abnormalities that could reasonably be expected to occur in the course of system operation. The principles presented are applicable to both new electrical system design and to the changing, upgrading, or expansion of an existing electrical distribution system.

[1]Information on references can be found in 1.8.

1.1.2 Objectives

The objectives of electrical system protection and coordination are to

— Limit the extent and duration of service interruption whenever equipment failure, human error, or adverse natural events occur on any portion of the system
— Minimize damage to the system components involved in the failure

The circumstances causing system malfunction are usually unpredictable; however, sound design and preventive maintenance can reduce the likelihood of system problems. The electrical system, therefore, should be designed and maintained to protect itself automatically.

1.1.2.1 Safety

Prevention of human injury is the most important objective when designing electrical systems. Interrupting devices should have adequate interrupting capability. Energized parts should be sufficiently enclosed or isolated to avoid exposing personnel to explosion, fire, arcing, or shock. Safety should always take priority over service continuity, equipment damage, or economics.

These fundamental principles of safety have always been adhered to by responsible engineers engaged in the design and operation of electrical systems. The National Electrical Code® (NEC®) (NFPA 70-1999), National Electrical Safety Code® (NESC®) (Accredited Standards Committee C2-2002), NFPA 70E-2000, and state and local codes all have prescribed practices intended to enhance the safety of electrical systems. In recent years, an increased concern about safety has led to many studies resulting in detailed recommendations [from the National Institute of Occupational Safety and Health (NIOSH)] and regulations relating to electrical systems. Prominent among these are the regulations of the Occupational Safety and Health Administration (OSHA) of the U.S. Department of Labor. Engineers and other personnel engaged in the design and operation of electrical system protection should be familiar with the most recent OSHA regulations, NIOSH Safely Alerts, and all other applicable codes and regulations relating to human safety.

The essential electrical infrastructure in industrial and commercial establishments employs protective devices, many of which are addressed in this recommended practice, that function to de-energize the electrical system in the event of a malfunction. However, electric shock can result in serious injury far more quickly than the available technology of interruption can perform its task. Therefore, the limitation of shock hazard, such as the step and touch potentials in electrical substations, depends upon the speed of interrupting the fault. Rapid fault isolation is important (see IEEE Std 80-2000). Therefore, the most efficient form of electrical shock protection is to avoid shock altogether, and such avoidance is best accomplished through proper system design and operation, and through effective maintenance.

1.1.2.2 Equipment damage versus service continuity

Whether minimizing the risk of equipment damage or preserving service continuity is the more important objective depends upon the operating philosophy of the particular industrial plant or commercial business. Some operations can afford limited service interruptions to minimize the possibility of equipment repair or replacement costs, while others would regard such an expense as small compared with even a brief interruption of service.

In most cases, electrical protection should be designed for the best compromise between equipment damage and service continuity. One of the prime objectives of system protection is to obtain selectivity to minimize the extent of equipment shutdown in case of a fault. Therefore, many protection engineers would prefer that faulted equipment be de-energized as soon as the fault is detected.

However, for certain continuous process industry plants, high-resistance grounding systems that allow the first ground fault to be alarmed instead of automatically cleared are employed. These systems are described in Chapter 8.

1.1.2.3 Economic and reliability considerations

The cost of system protection determines the degree of protection that can be feasibly designed into a system. Many features may be added that improve system performance, reliability, and flexibility, but incur an increased initial cost. However, failure to design into a system at least the minimum safety and reliability requirements inevitably results in unsatisfactory performance, with a higher probability of expensive downtime. Modifying a system that proves inadequate is more expensive and, in most cases, less satisfactory than initially designing these features into a system.

The system should always be designed to isolate faults with a minimum disturbance to the system and should have features to give the maximum dependability consistent with the plant requirements and justifiable costs. Evaluation of costs should also include equipment maintenance requirements. In many instances, plant requirements make planned system outages for routine maintenance difficult to schedule. Such factors should weigh into the economic-versus-reliability decision process.

When costs of downtime and equipment maintenance are factored into the protection system cost evaluation, decisions can then be based upon total cost over the useful life of the equipment rather than simply the first cost of the system. In-depth coverage of reliability-versus-economic decisions can be found in IEEE Std 493-1997.

1.2 Protection against abnormalities

The principal electrical system abnormalities to protect against are short circuits and overloads. Short circuits may be caused in many ways, including failure of insulation due to excessive heat or moisture, mechanical damage to electrical distribution equipment, and failure of utilization equipment as a result of overloading or other abuse. Circuits may become overloaded simply by connecting larger or additional utilization equipment to the circuit. Overloads may also be caused by improper installation and maintenance, such as misaligned shafts and worn bearings. Improper operating procedures (e.g., too frequent starting, extended acceleration periods, obstructed ventilation) are also a cause of equipment overload or damage.

Short circuits may occur between two-phase conductors, between all phases of a polyphase system, or between one or more phase conductors and ground. The short circuit may be solid (or bolted) or welded, in which case the short circuit is permanent and has a relatively low impedance. The extreme case develops when a miswired installation is not checked prior to circuit energization. In some cases the short circuit may burn itself clear, probably opening one or more conductors in the process. The short circuit may also involve an arc having relatively high impedance. Such an arcing short circuit can do extensive damage over time without producing exceptionally high current. An arcing short circuit may or may not extinguish itself. Another type of short circuit is one with a high-impedance path, such as dust accumulated on an insulator, in which a flashover occurs. The flashover may be harmlessly extinguished or the ionization produced by the arc may lead to a more extensive short circuit. These different types of short circuits produce somewhat different conditions in the system.

Electrical systems should be protected against the highest short-circuit currents that can occur; however, this maximum fault protection may not simultaneously provide adequate protection against lower current faults, which may involve an arc that is potentially destructive.

Ground faults comprise the majority of all faults that occur in industrial and commercial power systems. Ground-fault currents may be destructive, even though the magnitude may be reduced by a high impedance in the fault and return path. Several methods of grounding are available; and the appropriate selection for the particular voltage level, combined with proper detection and relaying, can help achieve the goals of reduced damage and service continuity. For this reason, Chapter 8 is devoted to ground-fault protection.

With the increasing use of nonlinear system loads and devices, harmonics have become an ever-increasing system abnormality to contend with. Electrical system design, whether new or through changes or additions to an existing system, should take into account the possible effects of harmonic current and voltages on system equipment and protective devices. In many cases, harmonics may cause excessive heating in system components; improper operation of control, metering, and protective devices; and other problems.

Other sources of abnormality, such as lightning, load surges, and loss of synchronism, usually have little or no effect on system overcurrent selectivity, but should not be ignored. These abnormalities usually can be best handled on an individual protective basis for the specific equipment involved (e.g., transformers, motors, generators).

1.3 Planning system protection

The designer of electrical power systems has available several techniques to minimize the effects of abnormalities occurring on the system or in the utilization equipment that the system supplies. One can design into the electric system features that

a) Quickly isolate the affected portion of the system and, in this manner, maintain normal service for as much of the system as possible. This isolation also minimizes damage to the affected portion of the system.
b) Minimize the magnitude of the available short-circuit current and, in this manner, minimize potential damage to the system, its components, and the utilization equipment it supplies.
c) Provide alternate circuits, automatic transfers, or automatic reclosing devices, where applicable, in order to minimize the duration or the extent of supply and utilization equipment outages.

System protection encompasses all of the above techniques; however, this text deals mainly with the prompt isolation of the affected portion of the system. Accordingly, the function of system protection may be defined as the detection and prompt isolation of the affected portion of the system whenever a short-circuit or other abnormality occurs that might cause damage to, or adversely affect, the operation of any portion of the system or the load that it supplies.

Coordination is the selection and/or setting of protective devices in order to isolate only the portion of the system where the abnormality occurs. Coordination is a basic ingredient of a well-designed electrical distribution protection system and is mandatory in certain health care and continuous process industrial systems.

System protection is one of the most basic and essential features of an electrical system and should be considered concurrently with all other essential features. Too often system protection is considered after all other design features have been determined and the basic system design has been established. Such an approach may result in an unsatisfactory system that cannot be adequately protected, except by a disproportionately high expenditure. The designer should thoroughly examine the question of system protection at each stage of planning and incorporate into the final system a fully integrated protection plan that is capable of, and is flexible enough, to grow with the system.

In planning electric power systems, the designer should endeavor to keep the final design as simple as would be compatible with safety, reliability, maintainability, and economic considerations. Designing additional reliability or flexibility into a system may lead to a more complex system requiring more complex coordination and maintenance of the protective system. Such additional complexity should be avoided except where the requisite personnel, equipment, and know-how are available to adequately service and maintain a complex electric power system.

1.4 Preliminary design

The designer of an electrical power system should first determine the load requirement, including the sizes and types of loads, and any special requirements. The designer should also determine the available short-circuit current at the point of delivery, the time-current curves and settings of the nearest utility protective devices, and any contract restrictions on ratings and settings of protective relays or other overcurrent protective devices in the user's system.

(See Figure 1-1.) The designer can then proceed with a preliminary system design and preparation of a one-line diagram.

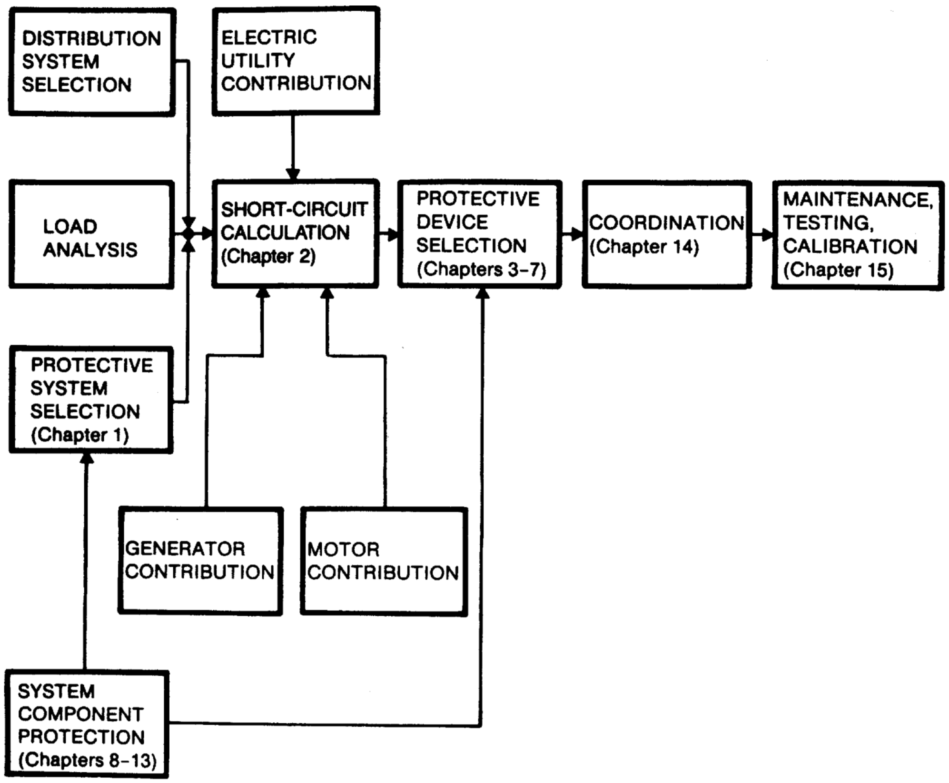

Figure 1-1—Sequence of steps in system protection and coordination

Chapter 2 covers the fundamentals of short-circuit analysis and the calculation of short-circuit duty requirements that permit evaluation of the preliminary design for compatibility with available ratings of circuit breakers, fuses, and other devices. At this point, some modification of the design may be necessary because of economic considerations or equipment availability, or both.

The preliminary design should be evaluated from the standpoint of system coordination, as covered in Chapter 15. If the protective devices provided in the preliminary design cannot be selectively coordinated with utility protective device settings and contractual restrictions on protective device settings, the design should be modified to provide proper selective coordination.

Ground-fault protection is an essential part of system protection and is given detailed coverage in Chapter 8 for two reasons. First, although many of the devices used to obtain ground-fault protection are similar to those covered in Chapter 3 through Chapter 7, the need for such protection and the potential problems of improper application of ground-fault protection are frequently not fully appreciated. Second, proper selective coordination of ground-fault protection seldom causes any change to overall system selectivity, although its effect should be taken into consideration in the same general manner covered in Chapter 15. In-depth coverage of system grounding can also be found in IEEE Std 142-1991.

Throughout the preliminary and final design process, the personal computer has become an indispensable tool in power system planning, analysis, and simulation of day-to-day operations. A number of power system software programs are available to assist the designer in evaluating protective device application and to assist in the proper selection and coordination of protective devices. Available software includes programs to evaluate and perform short circuit, protective device coordination, load flow, harmonic analysis, system stability, motor starting, and grounding. In-depth coverage of power system analysis can be found in IEEE Std 399-1997.

The designer of the protective system should bear in mind that the design consumes two critical and limited resources, space and money, and should take practical steps to ensure that these needs are fully recognized by the overall project team.

1.5 Basic protective equipment

The isolation of short circuits and overloads requires the application of protective equipment that senses when an abnormal current flow exists and then removes the affected portion from the system.

The three primary protective equipment components used in the isolation of short circuits and overloads are fuses, circuit breakers, and protective relays.

A fuse is both a sensing and interrupting device, but not a switching device. It is connected in series with the circuit and responds to thermal effects produced by the current flowing through it. The fusible element is designed to open at a predetermined time depending upon the amount of current that flows. Different types of fuses are available having time-current characteristics required for the proper protection of the circuit components. Fuses may be noncurrent-limiting or current-limiting, depending upon their design and construction. Fuses are not resetable because their fusible elements are consumed in the process of interrupting the current flow. Fuses and their characteristics, applications, and limitations are described in detail in Chapter 5 and Chapter 6.

Circuit breakers are interrupting and switching devices that require overcurrent elements to fulfill the detection function. In the case of medium-voltage (1–72.5 kV) circuit breakers, the sensing devices are separate current transformers (CTs) and protective relays or combinations of relays. These devices are covered in Chapter 4. For most low-voltage (under 1000 V) circuit breakers, (molded-case circuit breakers or low-voltage power circuit breakers) the sensing elements are an integral part of the circuit breaker. These trip units may be thermal or magnetic series devices; or they may be integrally mounted, but otherwise separate electronic devices used with CTs mounted in the circuit breaker. Low-voltage circuit breakers, their applications, characteristics, and limitations are covered in Chapter 7.

Overcurrent relays used in conjunction with medium-voltage circuit breakers are available with a range of different functional characteristics. Relays may be either directional or nondirectional in their action. Relays may be instantaneous and/or time-delay in response. Various time-current characteristics (e.g., inverse time, very inverse time, extremely inverse time, definite minimum time) are available over a wide range of current settings. Overcurrent relays and their selection, application, and settings are covered in detail in Chapter 4. Numerous other types of protective relays, used for specific protective purposes, are also discussed throughout this publication. Relays generally are used in conjunction with instrument transformers, which are covered in Chapter 3.

1.6 Special protection

In addition to developing a basic protection design, the designer may also need to develop protective schemes for specific equipment or for specific portions of the system. Such specialized protection should be coordinated with the basic system protection. Specialized protection applications include

— Conductor protection (see Chapter 9)
— Motor protection (see Chapter 10)
— Transformer protection (see Chapter 11)
— Generator protection (see Chapter 12)
— Bus and switchgear protection (see Chapter 13)
— Service supply-line protection (see Chapter 14)

1.7 Field follow-up

Proper application of the principles covered in the first 15 chapters of this recommended practice should result in the installation of system protection capable of coordinated selective isolation of system faults, overloads, and other system problems. However, this capability will be useless if the proper field follow-up is not planned and executed. Field follow-up has three elements: proper installation, including testing and calibration of all protective devices; proper operation of the system and its components; and a proper preventive maintenance program, including periodic retesting and recalibration of all protective devices. A separate chapter, Chapter 16, has been included to cover testing and maintenance.

1.8 References

This recommended practice shall be used in conjunction with the following standards. When the following standards are superseded by an approved revision, the revision shall apply.

ASC C2-2002, National Electric Safety Code® (NESC®).[2]

IEEE Std 80-2000, IEEE Guide for Safety in AC Substation Grounding.[3]

IEEE Std 141-1993 (Reaff 1999), IEEE Recommended Practice for Electric Power Distribution for Industrial Plants *(IEEE Red Book*).

IEEE Std 142-1991, IEEE Recommended Practice for Grounding of Industrial and Commercial Power Systems *(IEEE Green Book*).

IEEE Std 241-1990 (Reaff 1997), IEEE Recommended Practice for Electric Power Systems in Commercial Buildings *(IEEE Gray Book*).

IEEE Std 399-1997, IEEE Recommended Practice for Industrial and Commercial Power Systems Analysis (*IEEE Brown Book*).

IEEE Std 493-1997, IEEE Recommended Practice for the Design of Reliable Industrial and Commercial Power Systems (*IEEE Gold Book*).

NFPA 70-1999, National Electrical Code® (NEC®).[4]

NFPA 70E-2000, Electrical Safety Requirements for Employee Workplaces.[5]

[2]The NESC is available from the Institute of Electrical and Electronics Engineers, 445 Hoes Lane, P.O. Box 1331, Piscataway, NJ 08855-1331, USA (http://standards.ieee.org/).

[3]IEEE publications are available from the Institute of Electrical and Electronics Engineers, 445 Hoes Lane, P.O. Box 1331, Piscataway, NJ 08855-1331, USA (http://standards.ieee.org/).

[4]The NEC is published by the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269, USA (http://www.nfpa.org/). Copies are also available from the Institute of Electrical and Electronics Engineers, 445 Hoes Lane, P.O. Box 1331, Piscataway, NJ 08855-1331, USA (http://standards.ieee.org/).

[5]NFPA publications are published by the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269, USA (http://www.nfpa.org/).

Chapter 2
Short-circuit calculations

2.1 Introduction

Short-circuit currents can create massive destruction to the power system. Short circuits typically have magnitudes many times greater than load currents. The consequences of these high-magnitude currents can be catastrophic to normal operation of the power system. First, the presence of short-circuit currents in system conductors results in additional heating, which the system is usually not designed to sustain continuously. These currents also introduce severe mechanical forces on conductors, which can break insulators, distort transformer windings, or cause other physical damage. The flow of high-magnitude short-circuit currents through system impedances may also result in abnormally low voltages, which in turn lead to otherwise healthy equipment being forced to shut down. Finally, at the point of the short circuit itself, generally the release of energy in the form of an arc, if left uncorrected, can start a fire, which may spread well beyond the point of initiation.

Much of the effort of power system engineers and planners is directed toward minimizing the impact of short circuits on system components and the industrial process the system serves. It has been said that the only part of a power system that actually works is the protective devices that are called upon to detect and react to short circuits—and they only do something when something else goes wrong! This has led to the observation that power system engineers focus only on catastrophes. It is true that a great deal of power system engineering is devoted to the analysis of undesirable events, and nowhere is that statement more correct than in the application of protective devices.

Lord Kelvin observed that knowledge of a subject is incomplete until the subject can be accurately quantified. Of all the demands placed upon the power system protection engineer, the most analytical is to determine the magnitude of voltages and currents that the system can produce under various short-circuit conditions. Only when these quantities are understood can the application of protective devices proceed with confidence that they will perform their intended function when short circuits occur.

The most fundamental principle involved in determining the magnitude of short-circuit current is Ohm's Law: the current that flows in a network of impedances is related to the driving voltage by the relationship

$$I = \frac{E}{Z} \tag{2-1}$$

The general procedure for applying this principle entails the three steps involved in Thevenin's Theorem of circuits.

a) Develop a graphical representation of the system, called a one-line (or single-line) diagram, with symbolic voltage sources and circuit impedances.

b) Calculate the total impedance from the source of current (i.e., the driving voltage) to the point at which a hypothetical short-circuit current is to be calculated. This value is the Thevenin equivalent impedance, sometimes called the driving point impedance.
c) Knowing the open circuit prefault voltage, use Ohm's Law to calculate the short-circuit current magnitude.

Of course, the actual application of these basic principles is more involved, and the remainder of this chapter is devoted to a treatment of the specific details of short-circuit current calculations.

2.2 Types of short-circuit currents

From the point of view of functional application, four or more distinct types of short-circuit current magnitudes exist. The current of greatest concern flows in the system under actual short-circuit conditions and could (at least theoretically) be measured using some form of instrumentation. In reality, it is not practical to attempt to predict by calculation the magnitude of actual current because it is subject to a great many uncontrollable variables. Power system engineers have developed application practices, some of which are discussed in the following paragraphs, that predict worst-case magnitudes of current sufficient for application requirements.

The analyst or engineer may have several objectives in mind when a short-circuit current magnitude is calculated. Obviously, the worst-case current should be appropriate to the objective, and a set of assumptions that leads to a worst-case calculation for one purpose may not yield worst-case results for another purpose.

Short-circuit current magnitudes often must be calculated in order to assess the application of fuses, circuit breakers, and other interrupting devices relative to their ratings. These currents have labels (e.g., interrupting duty, momentary duty, close and latch duty, breaking duty), which correlate those magnitudes with the specific interrupter rating values against which they should be compared to determine whether the interrupting device has sufficient ratings for the application. ANSI standard application guides define specific procedures for calculating duty currents for evaluating fuses and circuit breakers rated under ANSI standards. Likewise, the International Electrotechnical Commission (IEC) publishes a calculation guide for calculating duty currents for IEC-rated interrupting devices. In either case, the important thing is that the basis for calculating the current be consistent with the basis for the device rating current so that the comparison is truly valid.

Related to interrupter rating currents are the currents used to evaluate the application of current-carrying components. Transformers, for example, are designed to have a fault withstand capability defined in terms of current, and transformer applications should be evaluated to assure that these thermal and mechanical limitations are being observed. Likewise, bus structures should be designed structurally to withstand the forces associated with short circuits, and this requires knowledge of the magnitude of available fault currents. Similarly, ground grids under electrical structures should be designed to dissipate fault currents without causing excessive voltage gradients. In each case, it is necessary to calculate a fault magnitude in a fashion that is consistent with the purpose for which it is needed.

Another type of short-circuit current magnitude is used by protection engineers to assess the time-current performance of protective devices. Here, again, consistency is needed between the calculated currents and the currents that the protective devices measure. No universally accepted standards define how protective devices make measurements, and in fact measurable differences exist among manufacturers, among technologies, and even among design vintages of the same manufacturer and technology. However, protection engineers have evolved a series of generally accepted guidelines for which currents apply to which kinds of protective devices, and these guidelines are detailed in subsequent chapters.

Other references in the *IEEE Color Book Series*™ treat the application of interrupting devices. Accordingly, this chapter discusses only the calculation of short-circuit currents for evaluating the time-current performance of relays, fuses, low-voltage circuit breaker trip devices, and other protective equipment.

Another way of looking at short circuits is to consider the geometry of faults. Most modern power systems are three-phase and involve three power-carrying conductors. A fourth conductor, the neutral, may or may not carry load current depending upon the nature of the loads on the system. The number of conductors involved in the short circuit has a bearing on the severity of the fault as measured by the magnitude of short-circuit current; normally, a fault involving all three-phase conductors (called the three-phase fault) is considered the most severe. Other geometries include single phase-to-ground faults, phase-to-phase faults, double phase-to-ground faults, and open conductors.

2.3 The nature of short-circuit currents

Under normal system conditions, the equivalent circuit of Figure 2-1 may be used to calculate load currents. Three impedances determine the flow of current. Z_s and Z_c are the impedances of the source and circuit, respectively, while Z_l is the impedance of the load. The load impedance is generally the largest of the three, and it is the principle determinant of the current magnitude. Load impedance is also predominantly resistive, with the result that load current tends to be nearly in phase with the driving voltage.

A short circuit may be thought of as a conductor that shorts some of the impedances in the network while leaving others unchanged. This situation is depicted in Figure 2-2. Because Z_s and Z_c become the only impedances that restrict the flow of current, the following observations may be made:

a) The short-circuit current is greater than load current.

b) Because Z_s and Z_c are predominately inductive, the short-circuit current lags the driving voltage by an angle approaching the theoretical maximum of 90°.

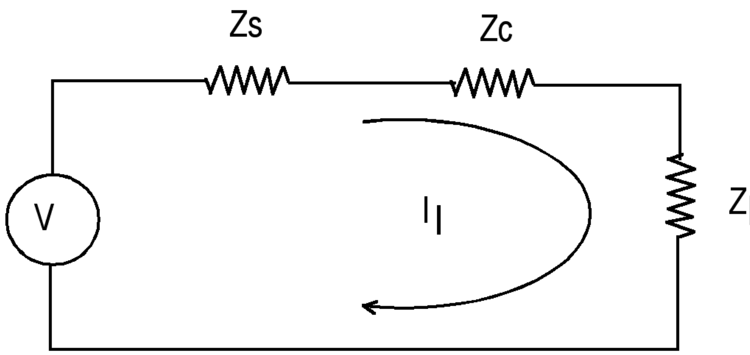

Figure 2-1—Equivalent circuit used to calculate load current in a normal circuit

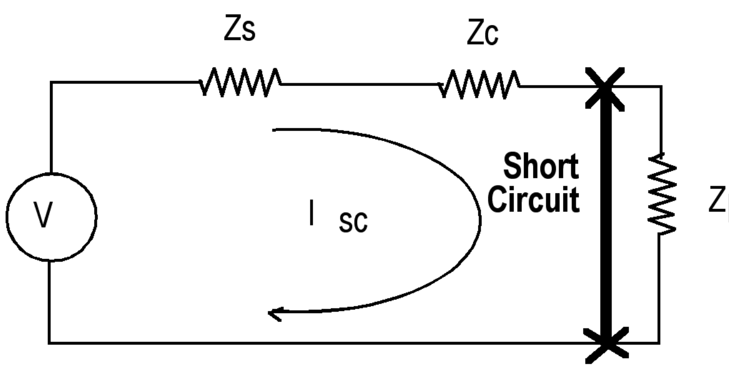

Figure 2-2—Equivalent circuit used to calculate short-circuit current

The change in state from load current to short-circuit current occurs rapidly. Fundamental physics demonstrate that the magnitude of current in an inductor cannot change instantaneously. This conflict can be resolved by considering the short-circuit current to consist of two components:

— A symmetrical ac current with the higher magnitude of the short-circuit current
— An offsetting dc transient with an initial magnitude that is equal to the initial value of the ac current, but which decays rapidly

The initial magnitude of the dc transient is directly controlled by the point on the voltage wave at which the short circuit occurs. If the short circuit occurs at the natural zero crossing of the driving voltage sinusoid, the transient is maximized. However, the transient is a minimum if the fault occurs at the crest of the voltage sinusoid. At any subsequent time, the magnitude of the dc transient is determined by the time constant of the decay of the dc, which is controlled by the ratio of reactance to resistance in the impedance limiting the fault. Equation (2-2) can be used to calculate the instantaneous magnitude of current at any time. For the protection engineer, the worst case initial current includes the full dc transient.

$$i(t) = \left(\frac{v}{Z_s + Z_c}\right)\left[\sin(wt + \varphi - \theta) - \sin(\varphi - \theta)e^{-\left(\frac{R}{L}\right)t}\right] \quad (2\text{-}2)$$

The driving voltage depicted in Figure 2-1 and Figure 2-2 is the Thevenin equivalent open-circuit voltage at the fault point prior to application of the short circuit. This voltage includes sources such as remote generators with voltage regulators that maintain their value regardless of the presence of a short circuit on the system as well as nearby sources whose voltages decay when the short circuit is present. The amount of decay is determined by the nature of the source. Nearby generators and synchronous motors with active excitation systems sustain some voltage, but because the short circuit causes their terminal voltage to drop, the current they produce is gradually reduced as the fault is allowed to persist. At the same time, induction motors initially participate as short-circuit current sources, but their voltages decay rapidly as the trapped flux is rapidly drained. Figure 2-3 shows the generic tendencies of various kinds of short-circuit current sources and a composite waveform for the symmetrical ac current decay. Figure 2-4 depicts the most realistic case of the decaying symmetrical ac current combined with the decaying dc transient. From this figure, a generalized short-circuit current may be described in the following terms:

— High initial magnitude dc transient component of current, which decays with time
— High initial magnitude symmetrical ac current, which diminishes gradually with time
— Symmetrical ac current lags driving voltage by a significant angle, approaching 90°

2.4 Protective device currents

It is the general practice to recognize three magnitudes of short-circuit current in applying protective devices. These magnitudes are fundamentally time-dependent and can be thought of as three points on the generic curve in Figure 2-4.

The first point is the initial magnitude and is considered by protection engineers to be the magnitude of current to which fast-acting protective devices respond. Instantaneous relays, fuses, and low-voltage circuit breaker trip devices are characterized as fast acting. In some instances this initial point includes the dc transient; in other instances, it does not. Whether the dc transient is recognized is determined by whether the protective device in question responds to dc quantities. For example, instantaneous relays operating on the induction principle, and static devices with dc filtering, respond only to the symmetrical ac component of this initial current, while fuses and plunger and hinged-armature relays sense the total magnitude of current. How the dc transient is treated is also subject to some interpretation. One traditional, conservative approach is to treat the initial current as though the magnitude

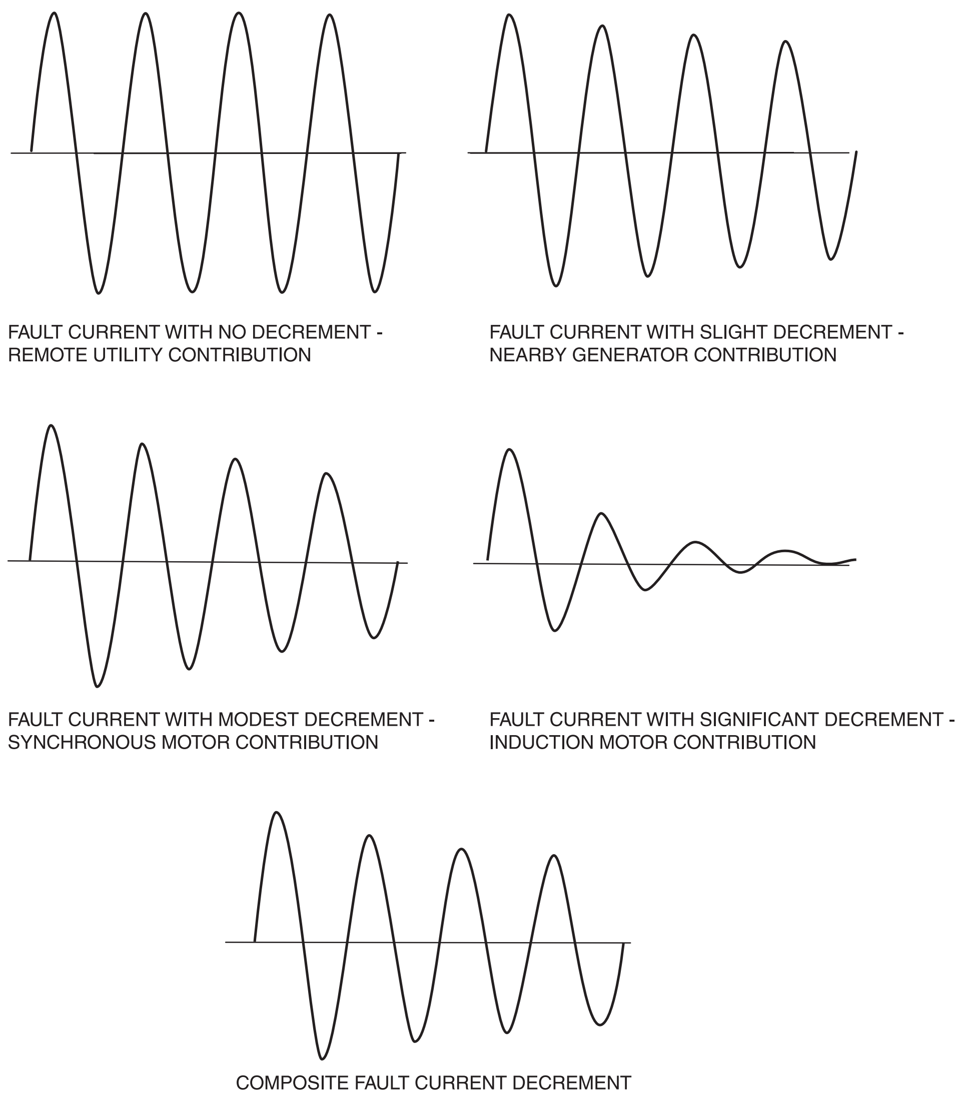

Figure 2-3—Generic components of fault current categorized according to decrement

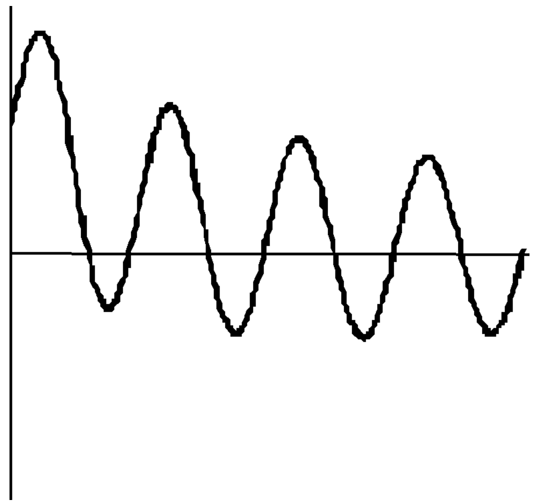

Figure 2-4—Typical fault current sinusoid with both ac and dc decrement

exists 0.5 cycle into the fault and, for typical systems where the *X/R* ratio is 25 or less, the root-mean-square (rms) total current does not exceed 1.6 times the symmetrical rms current. Some protection engineers choose instead to calculate an approximate asymmetry factor using the formula

$$\text{Asymmetry factor} = \sqrt{1 + 2e^{\frac{-2\pi}{X/R}}} \tag{2-3}$$

The second point is traditionally considered to be the magnitude of current at the time when time-delay overcurrent protective devices (e.g., overcurrent relays, delayed-action low-voltage trip devices) make their final measurement and operate. General practice assumes that the dc transient will have disappeared entirely by this time and to recognize only the rms symmetrical current.

The third current magnitude commonly calculated by protection engineers is the long-time current. Some protection engineers use the term “thirty (30) cycle current” because it is an estimate of the current that exists long after inception of the fault. This magnitude is used to evaluate performance of extremely long-time devices, such as generator backup overcurrent relays or second- or third-zone distance relays.

Determining the rates of decay of current to calculate these three time-based currents in exact form is difficult. The procedure that protection engineers have evolved is to represent the system using different impedances that result in short-circuit current magnitudes that are approximately close to the theoretically correct values. Table 2-1 summarizes common practices in this regard.

Table 2-1—Short-circuit impedances for protective device application and evaluation

Impedance	Instantaneous currents	Time-delay currents	Long-time currents
Remote system equivalent	R + **j**X	R + **j**X	R + **j**X
Local synchronous generators	R + **j**X"d	R + **j**X'd	R + **j**X
Synchronous motors	R + **j**X"d	R + **j**X'd	infinite
Induction motors	R + **j**X_{lr}	infinite	infinite
Passive components	R + **j**X	R + **j**X	R + **j**X

For conservatism, the usual practice is to employ saturated or rated-voltage reactances for rotating machines. These reactances are denoted by $_v$ on machine data sheets, e.g., $X\leq_{dv}$. Direct axis quantities are indicated by $_\mathrm{d}$, while " and ' indicate subtransient and transient values, respectively. Full data sheets are not always available for induction machines, and in this instance using locked-rotor reactance X_{lr} is common practice.

While including both the resistive and reactive components of impedance is classically correct, protection engineers sometimes employ the shortcut of ignoring resistance because the *X*/*R* ratio of impedances is typically quite high. This shortcut is especially common in higher voltage applications and when hand calculations are employed to arrive at answers quickly. Analysts are cautioned that in lower voltage systems, however, *X*/*R* ratios are low and to ignore resistance may lead to calculation of unacceptably high current magnitudes.

Earlier, the distinction was made between short-circuit currents calculated to evaluate the application of interrupting devices against their ratings and calculations needed to assess protective device performance. Many engineers bring these calculations together, using so-called momentary application calculations instead of performing a separate instantaneous calculation. Likewise, 5-cycle-to-8-cycle interrupting duty calculations are frequently used instead of doing a separate calculation of time-delay currents.

NOTE—Some of the impedances given in Table 2-1 differ slightly from prevailing practices employed in calculating short-circuit currents for interrupting device evaluations. Such discrepancies suggest that the calculations are not exact and room for judgment exists.

In addition to time considerations, short circuits vary by topography. In some instances, all three phases of the power system are involved in the short circuit while, under other conditions, the fault may consist of only one phase shorting to ground. It is possible for phase-to-phase faults to occur; and, in yet other instances, the phase-to-phase fault may also involve a current flow to ground. Any of these four geometric variants may or may not be bolted faults, that is, short circuits in which the conductors are shorted together with essentially no external impedance. In the real world, most faults involve some external impedance (and in fact arcing introduces external resistance), but protection engineers usually consider bolted faults as the worst case for determining fault current magnitude.

At times, determining the minimum currents available to system protective devices is necessary. No generally accepted procedure exists for determining minimum currents. One approach is to perform a system calculation considering only the electric utility source and any synchronous generators, with the latter represented by their synchronous reactance's (X_d), with allowance for presumed fault-point arcing represented by fault resistance.

In addition to short circuits, protection engineers may be called upon to evaluate performance of protective devices under other abnormal system conditions, such as open conductors.

2.5 Per-unit calculations

Power system calculations can be done using actual voltages and currents or using per-unit representations of actual quantities. While performing a calculation in actual quantities makes sense occasionally, the vast majority of calculations are done in per-unit. The discussion in this chapter assumes a familiarity with the per-unit method; but, to avoid confusion, definitions of important parameters are given in Table 2-2. The table presents strict (textbook) definitions and defines all per-unit values on a single-phase basis. Equivalent three-phase values are usually used in practice, but an understanding of the mathematics presented in Table 2-2 relies on a careful interpretation of base values as single-phase quantities.

Table 2-2—Per unit base parameters

Parameter	Strict definition (single-phase basis)	Common usage (three-phase basis)
Frequency	Steady state operating frequency	Nominal system frequency
Voltage	System line-to-neutral voltage at a chosen reference bus; voltage at other buses related by turns ratios of transformers	Nominal line-to-line voltage
MVA	Any defined arbitrary reference, per phase	10 or 100 MVA, 3ϕ
Base current	Base voltamperes divided by base voltage	Base 3ϕ kVA/(Line-to-line kV × $\sqrt{3}$)
Base impedance	Base line to neutral voltage divided by base current	(Line-to-line kV)^2/Base 3ϕ MVA

2.6 Short-circuit current calculation methods

2.6.1 Symmetrical components method of analysis

A variety of methods are commonly used for calculating short-circuit current magnitudes. All of these methods are ultimately traceable to the method of symmetrical components, and an

